

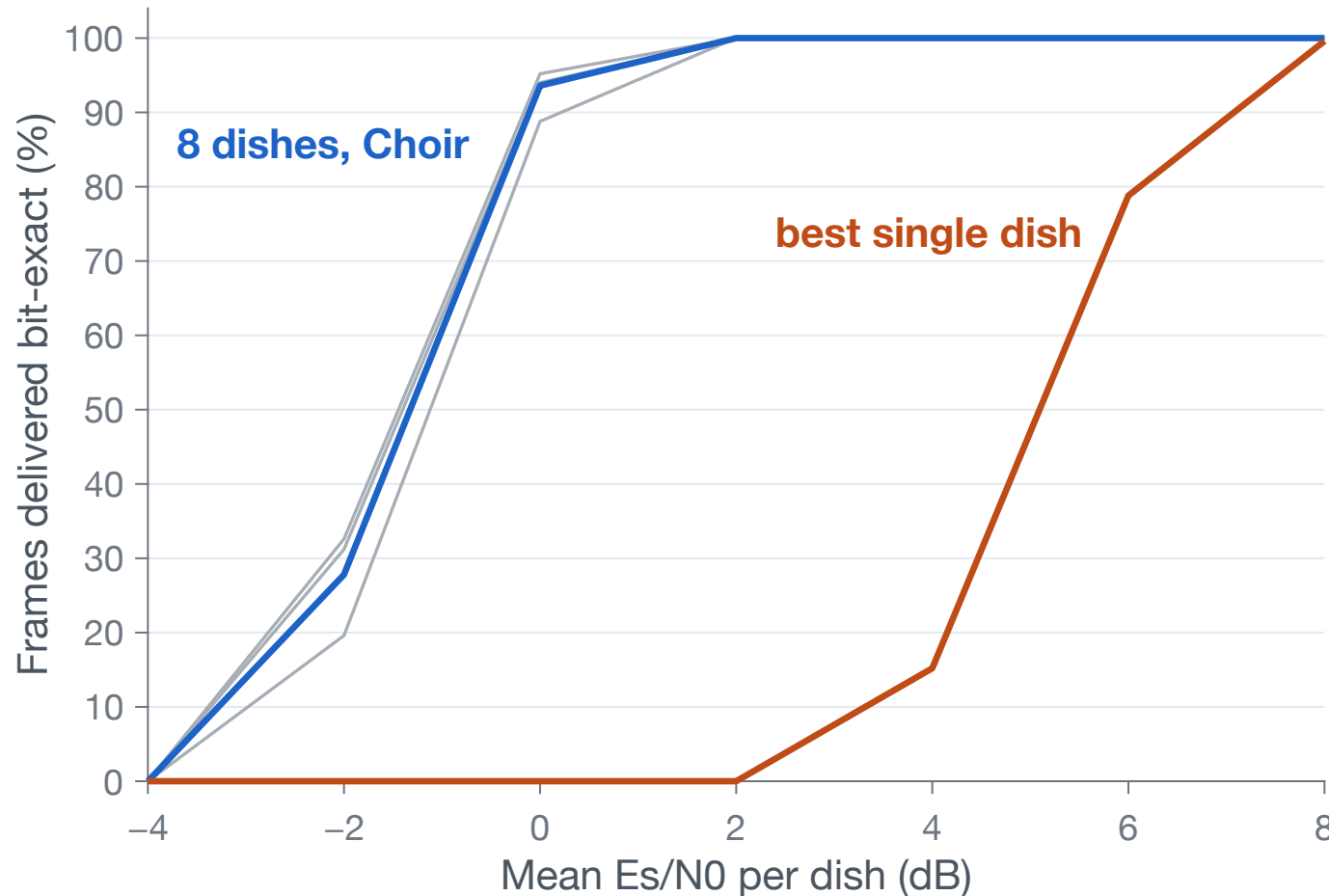
Choir keeps **88.5%** of telemetry frames when a look-alike spacecraft is 10 dB louder

An autonomous 8-dish receiver in MATLAB that learns which dishes to trust only from verified frames. Real NASA Curiosity (MSL) telemetry through a simulated radio link; JPL's blind SUMPLE method keeps 0%.

One idea, built twice: an 8-dish MATLAB receiver and a 3-LED breadboard that both trust what decodes

Part	What we built	What it shows
Track 1 brief	Find, clean and decode weak radio signals with no human tuning	Asked for: filter, acquire or re-acquire, decoded telemetry, before/after spectra
Software: Choir MATLAB R2026b	Real NASA Curiosity (MSL) telemetry in CCSDS frames, sent to 8 simulated dishes with hidden faults. Dish weights are learned only from frames that pass the CRC, carry our spacecraft ID and continue the frame count Telemetry is real; the radio link is simulated (no benchmark radio clips used)	Measured results on the next slides
Hardware: Choir on a breadboard Raspberry Pi Pico	3 LEDs take turns sending the same Morse frame (sync, ID C, counter, MATLAB, checksum) to one photoresistor. Each round a hidden fault garbles one LED, corrupts its payload, or puts a look-alike frame on 2 of 3	Live: raw vs filtered light signal, majority vote vs Choir's checks, decoded message in the terminal
Choir swarm add-on	Several receivers search carrier frequency, phase, timing and gain together (particle swarm), using a known training message	Last slide

One dish needs about 8 dB to decode a frame;
eight dishes decode every frame at **2 dB** each



99.6%

best single dish at 8 dB

15.2%

best single dish at 4 dB

100%

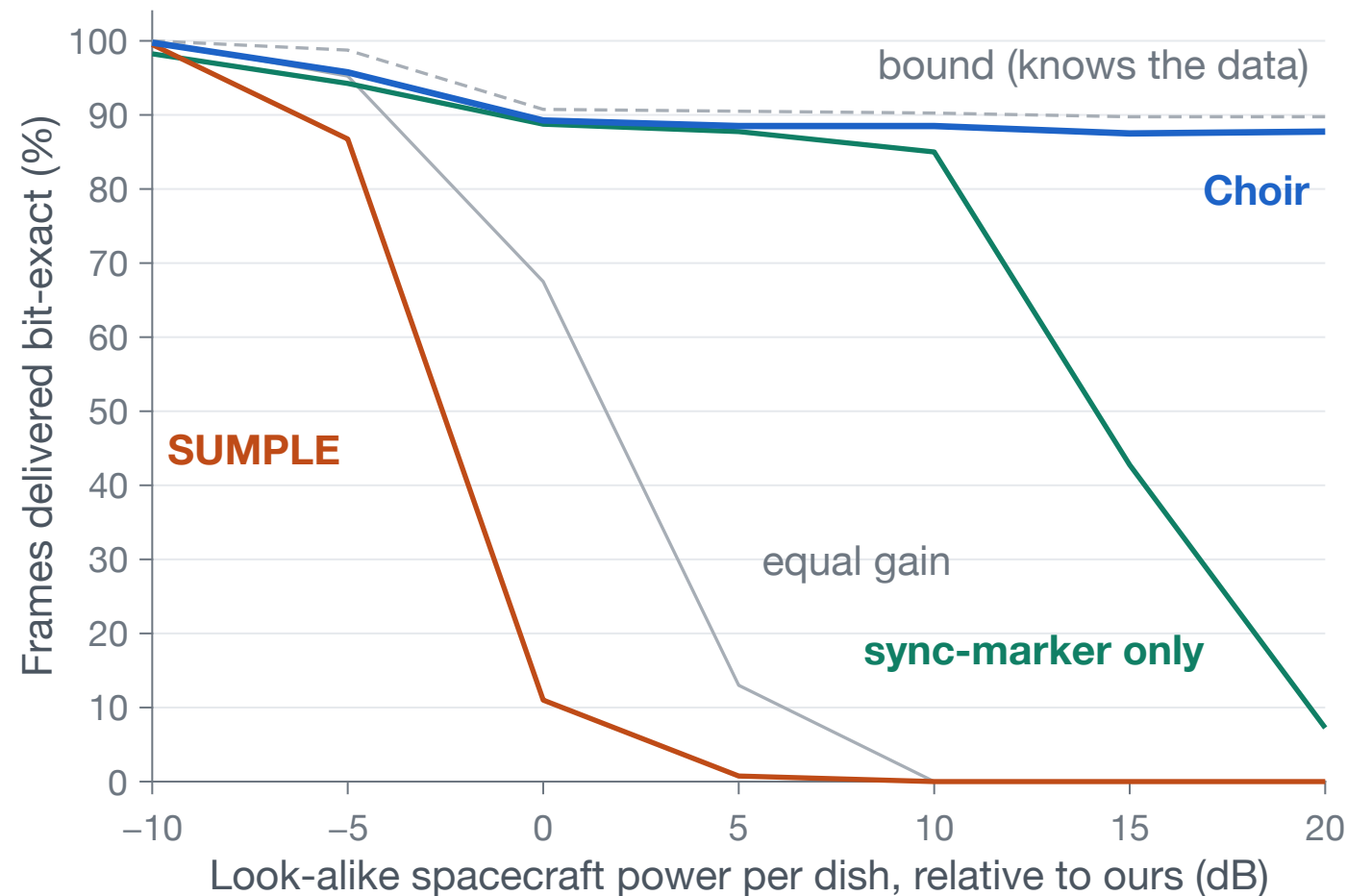
Choir, 8 dishes at 2 dB

Gray: the other 8-dish methods (equal gain, SUMPLE, sync-marker only). Without interference Choir matches them; its gain is robustness.

Choir runs itself in four states and learns dish weights only from frames that pass three checks

State	What Choir does	Moves on when
SEARCH	Squares the BPSK signal so the carrier shows as a tone; sums spectra over all 8 dishes	A peak stands 12× above the median
ALIGN	Finds the sync marker over one frame period on every dish: frame timing and per-dish delay Frames from another spacecraft ID are rejected	A frame passes all three checks
DECODE	Weights $w = Q^{-1}h$. h is learned only from frames with a valid CRC, our spacecraft ID 42 and the next frame count; Q is everything else the dishes hear A dish that stops matching verified frames is re-timed or excluded	3 bad frames in a row
FALLBACK	Re-times every dish on the sync marker; equal weights	A frame verifies, or 8 bad frames send it back to SEARCH

Against a look-alike spacecraft 10 dB louder,
Choir delivers **88.5%** of frames and SUMPLE 0%



85.0%

sync-marker only at +10 dB: Choir's math, trust from 32 known bits

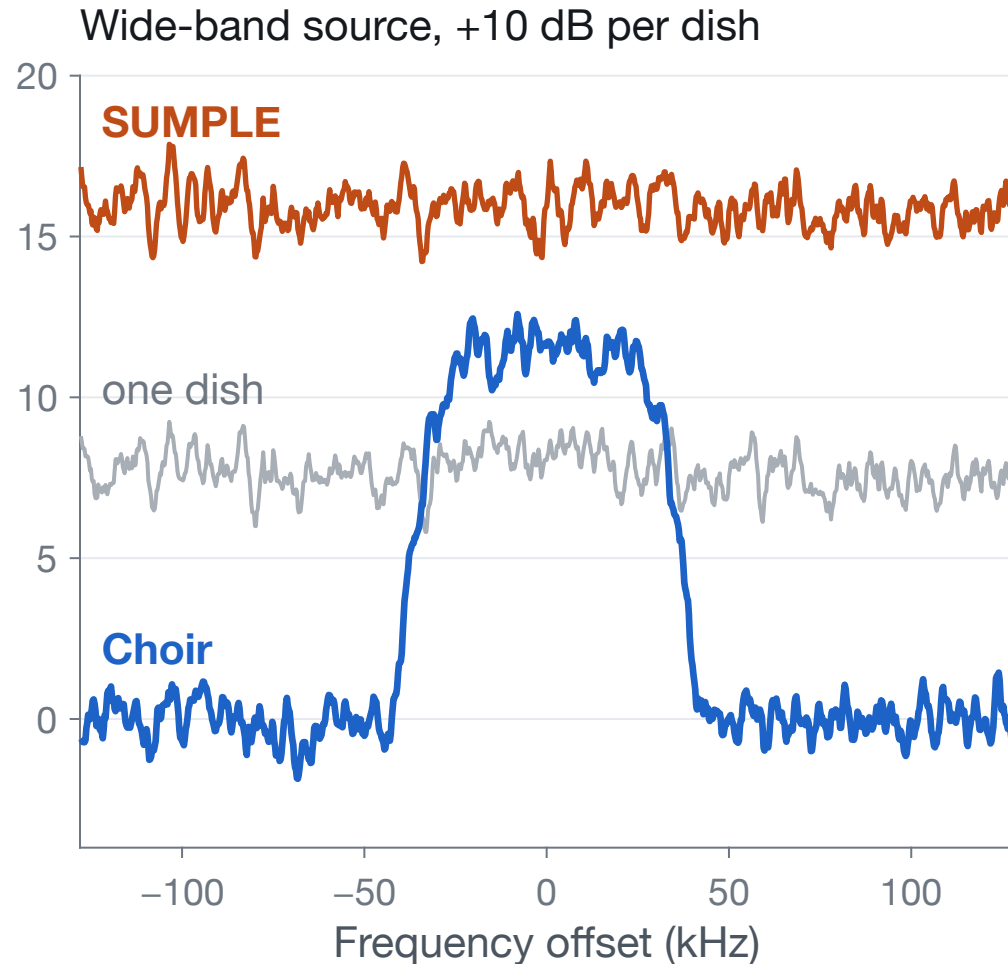
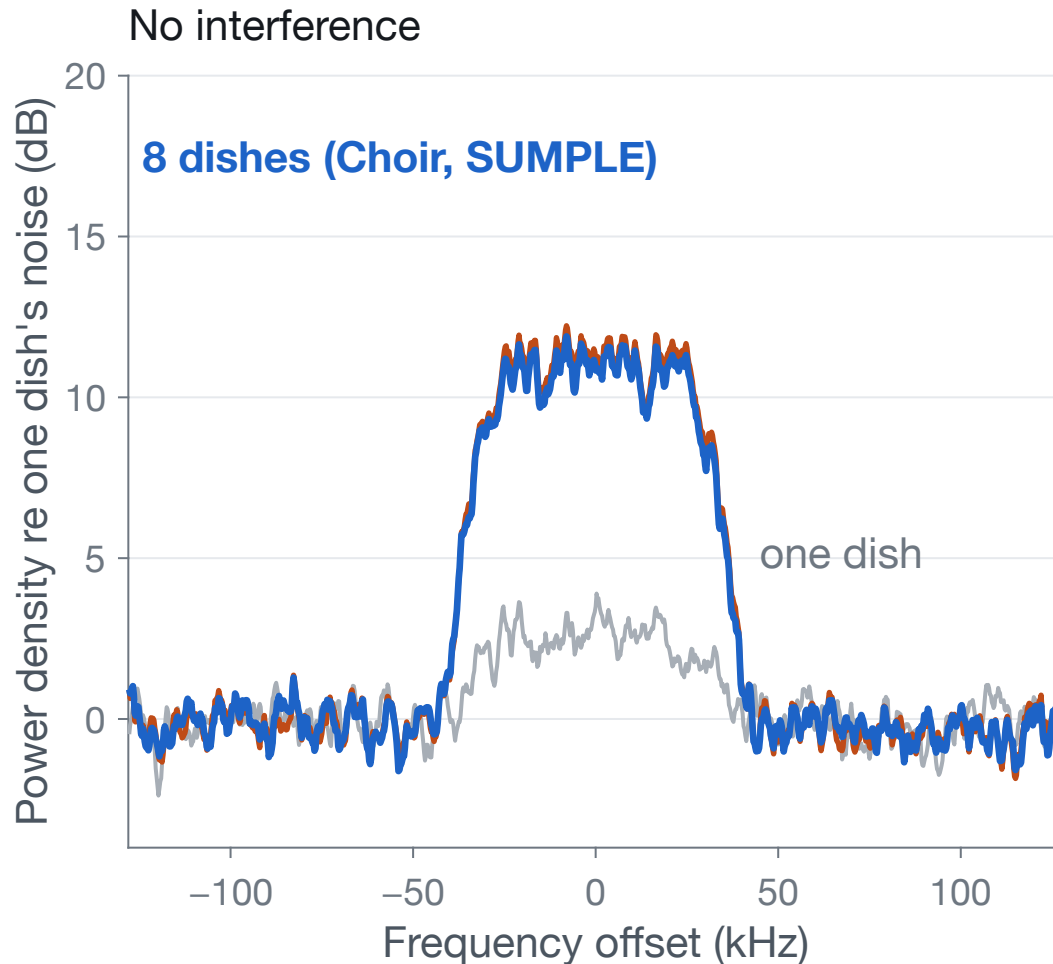
7.3% vs 87.8%

sync-marker only vs Choir at +20 dB

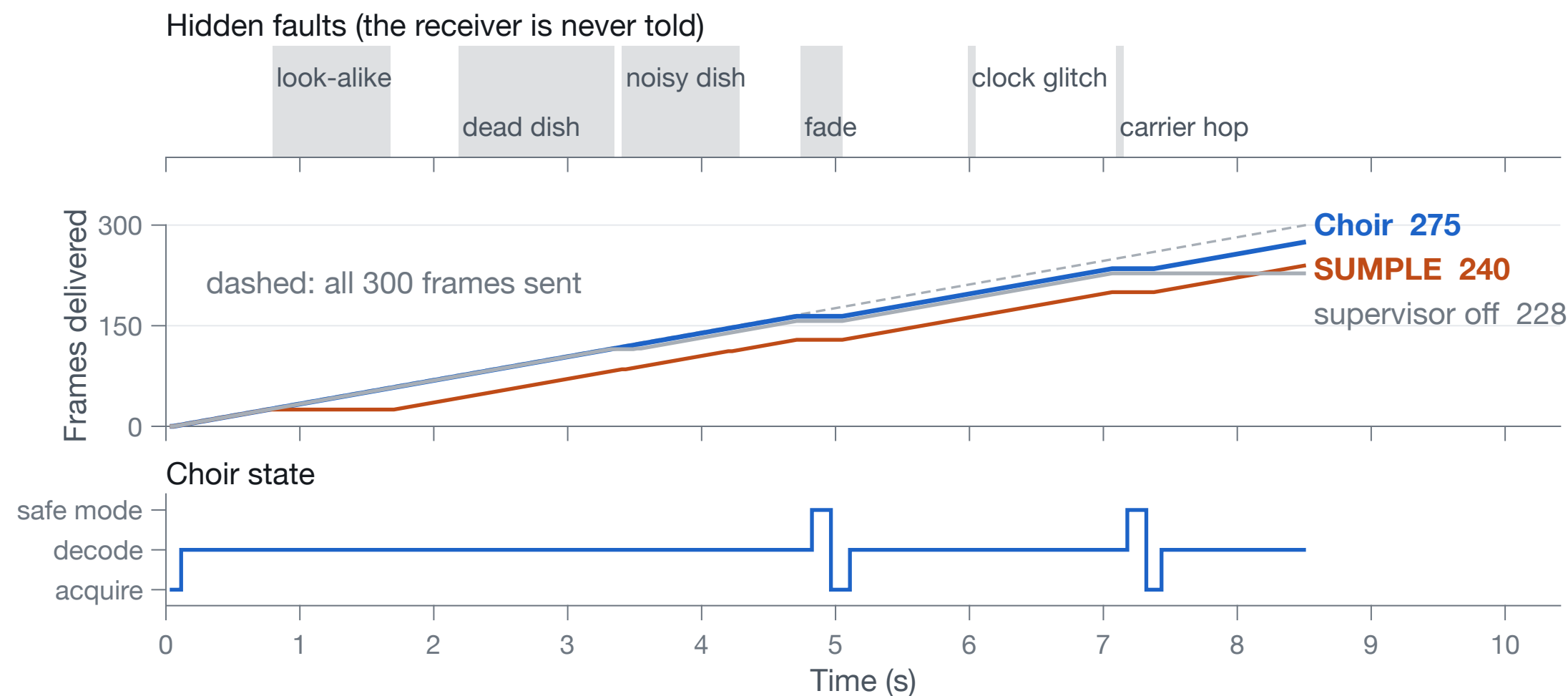
0

frames logged with wrong data, all interference runs

Under a +10 dB wide-band source, Choir holds the noise floor at **0.1 dB** while SUMPLE's rises to 16.0 dB



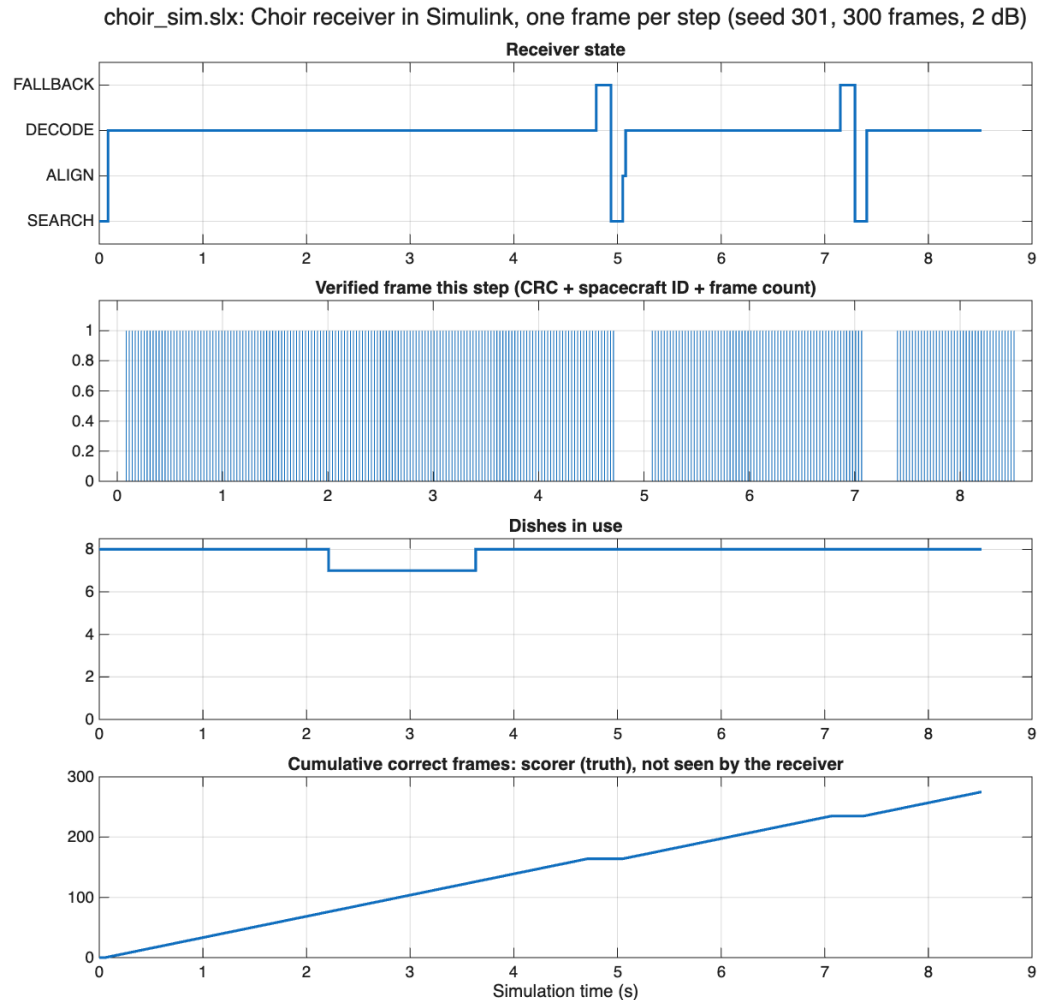
With six hidden faults per run, Choir delivers 90.6% of frames and is back 284 ms after a carrier hop



Choir logged each hidden fault within 2 frames (57 ms) of its start, with the reason

Hidden fault (truth)	Frame	Choir's log entry	Delay
Look-alike spacecraft, +10 dB	29	Strong extra source reaches 8 dishes (21 dB over noise); weights now cancel it	0
Dish 2 dead	78	Dish 2 stopped matching verified frames (fit 0.80 to 0.01): excluded	1
Dish 6 noise ×10	121	Dish 6 is 9 dB noisier than the floor; its weight drops	0
Fade, all dishes, −30 dB	168	3 bad frames in a row: every dish re-timed on the sync marker	2
Clock glitch, dish 1, −12 samples	212	Dish 1 stopped matching verified frames; timing moved −12 samples	0
Carrier hop, +1363 Hz	251	3 bad frames in a row: safe mode, then carrier found at +1440.5 Hz	2

The same receiver also runs inside Simulink, one frame per step, and delivers 275 of 300 frames as in MATLAB



On the breadboard, a look-alike frame on 2 of 3 LEDs is built to win the majority vote; Choir rejects it by its ID

Hidden fault (one per round)	What reaches the photoresistor	Majority vote trusts what agrees	Choir trusts what verifies
Noisy LED	One copy garbled by flicker	The 2 clean copies agree: correct	Garbled copy fails the checksum; a clean copy is logged
Corrupted payload	One copy with wrong letters but the old checksum	The 2 true copies agree: correct	That copy fails the checksum; a true copy is logged
Look-alike on 2 of 3 LEDs	Two copies with ID Q, payload DECOY and a valid checksum	DECOY wins 2 to 1: wrong	Both rejected (ID Q is not C); the true copy is logged

Choir fails where any array must: a look-alike arriving like our spacecraft (similarity 0.81) defeats every receiver

Limit	Measured effect	Next step
Look-alike from the same direction	Seed 304, similarity 0.81 (others 0.16–0.58): every receiver fails, even the bound that knows the data	Separate by frequency or code, not by dishes
No interference, weak signal	Choir 27.8% vs SUMPLE 32.6% at –2 dB per dish	Its gain is robustness, not sensitivity
Simulated radio link	Telemetry is real NASA MSL; dishes, noise and faults are simulated	Real multi-station recordings
Uncoded BPSK	One dish needs about 8 dB; LDPC or turbo codes need several dB less	Add coding; receiver logic unchanged

Swarm receivers learned each carrier offset to within 0.6 Hz and decoded two of three test messages with 0 bit errors

Message, Es/N0, receivers	True offset	Learned	True vs learned phase, timing	Decoded
HELLO FROM MARS, 8 dB, 8 receivers	40 Hz	39.5 Hz	30° vs 41.6°; 2.40 vs 2.95 samples	0 bit errors, 14 rounds
HI TEAM, 6 dB, 10 receivers	100 Hz	99.7 Hz	45° vs 48.6°; 4.00 vs 4.72 samples	0 bit errors, 10 rounds
DEEP SPACE LINK TEST, 4 dB, 12 receivers	−180 Hz	−179.4 Hz	120° vs 105°; 6.40 vs 6.06 samples	3 of 160 bits wrong; BPSK theory predicts about 2 at 4 dB